

DATA ANALYST INTERNSHIP

Task 2: Data Visualization and Storytelling

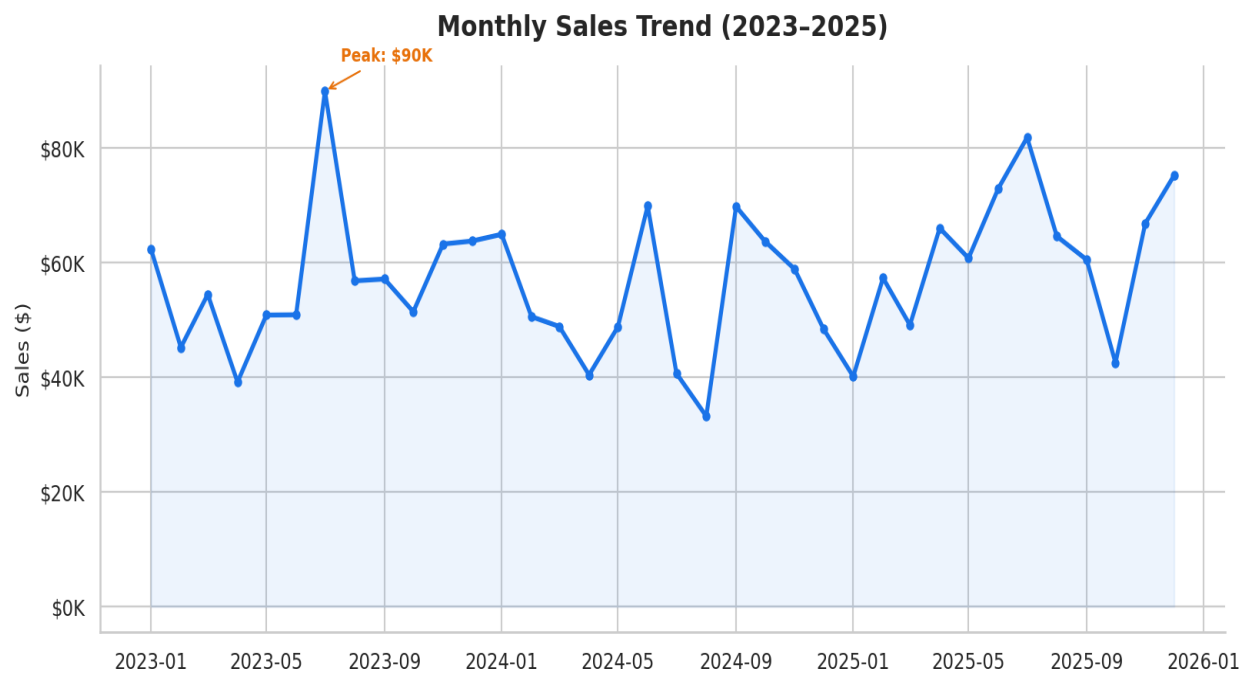
Objective: Create visualizations that convey a compelling business story from the Superstore sales dataset, highlighting trends, performance gaps, and actionable insights for decision-makers.

Dataset	Superstore Sales Data (1,850 orders, 2023–2025)
Tools Used	Python (Matplotlib, Seaborn, Pandas)
Deliverable	Visual Report (PDF) with 6 charts + storyboard summary
Date	30 June 2026

Executive Summary — Key Metrics

\$2.06M Total Sales	\$143.9K Total Profit	7.0% Profit Margin	1,850 Total Orders
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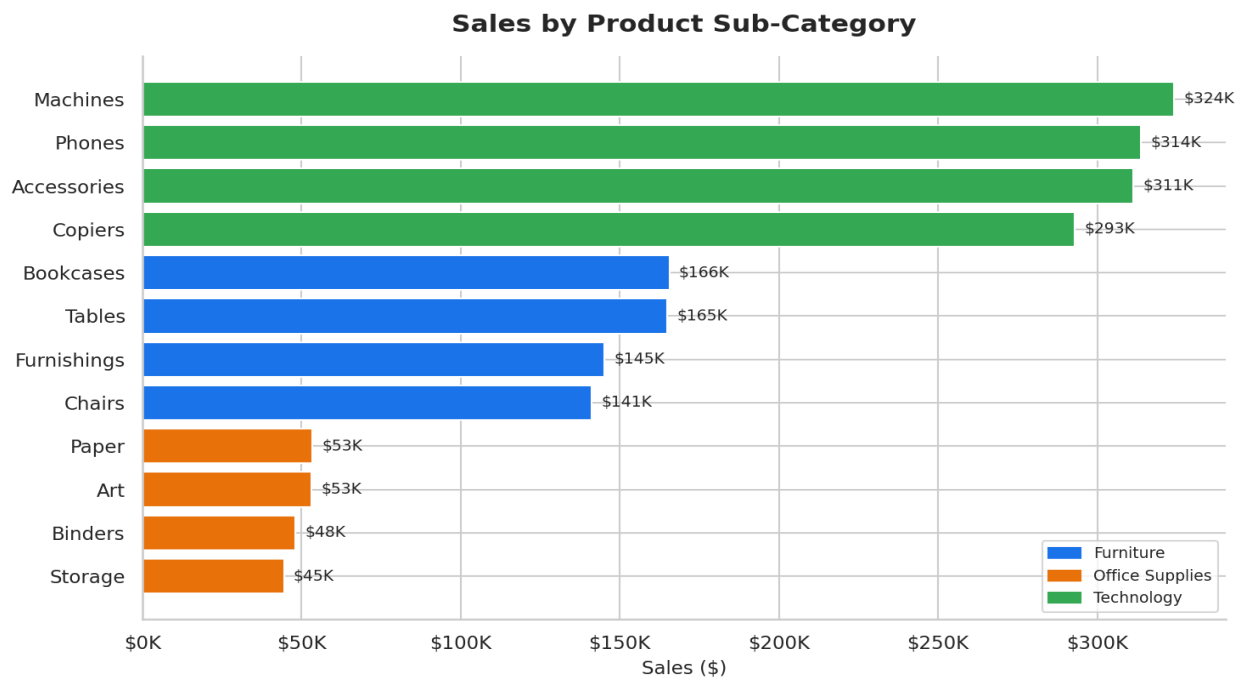
1. Monthly Sales Trend



■ Key Takeaway — Seasonal Growth Pattern

Sales show a clear upward trajectory from 2023 to 2025, with recurring peaks in Q4 (Nov–Dec), consistent with holiday shopping seasonality. This signals an opportunity to front-load inventory and marketing spend ahead of Q4 each year.

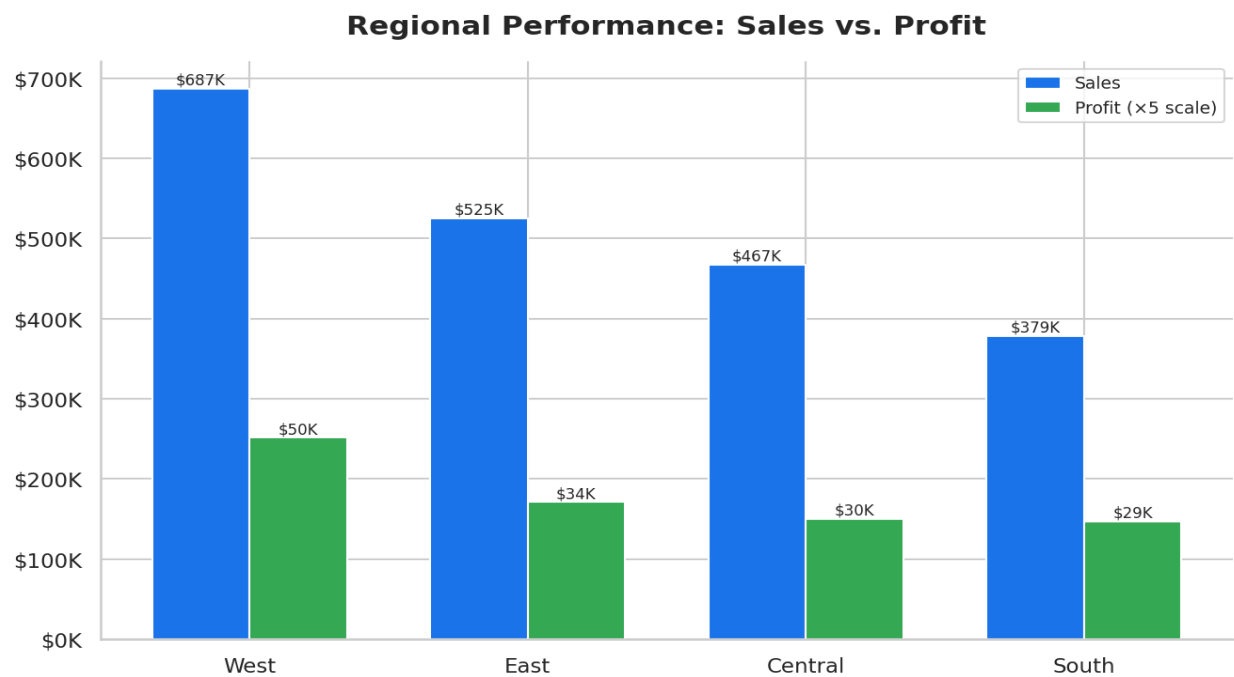
2. Sales by Product Sub-Category



■ Key Takeaway — Technology Leads Revenue

Machines and Phones (Technology) are the top revenue-generating sub-categories, while Furnishings and Art (Office Supplies) lag behind. Technology should remain the primary investment focus for promotions and stock allocation.

3. Regional Performance: Sales vs. Profit



■ **Key Takeaway — West Region Dominates**

The West region leads in both sales and profit, while the South region underperforms on both metrics. This gap suggests a need for region-specific strategies — potentially revisiting discount policy or marketing reach in the South.

4. Discount Impact on Profit

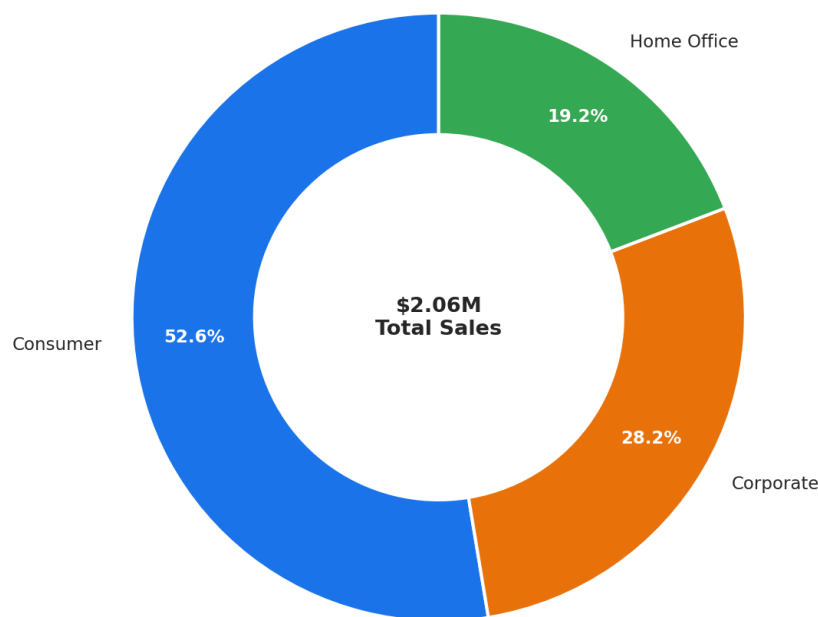


■ Key Takeaway — Discounts Above 30% Erode Profit

At 0% discount, average profit per order is \$192.96. Once discounts exceed 30-40%, average profit turns negative (-\$190.49). This is the single most actionable insight: capping discounts at 20-25% would protect margins significantly.

5. Sales Contribution by Customer Segment

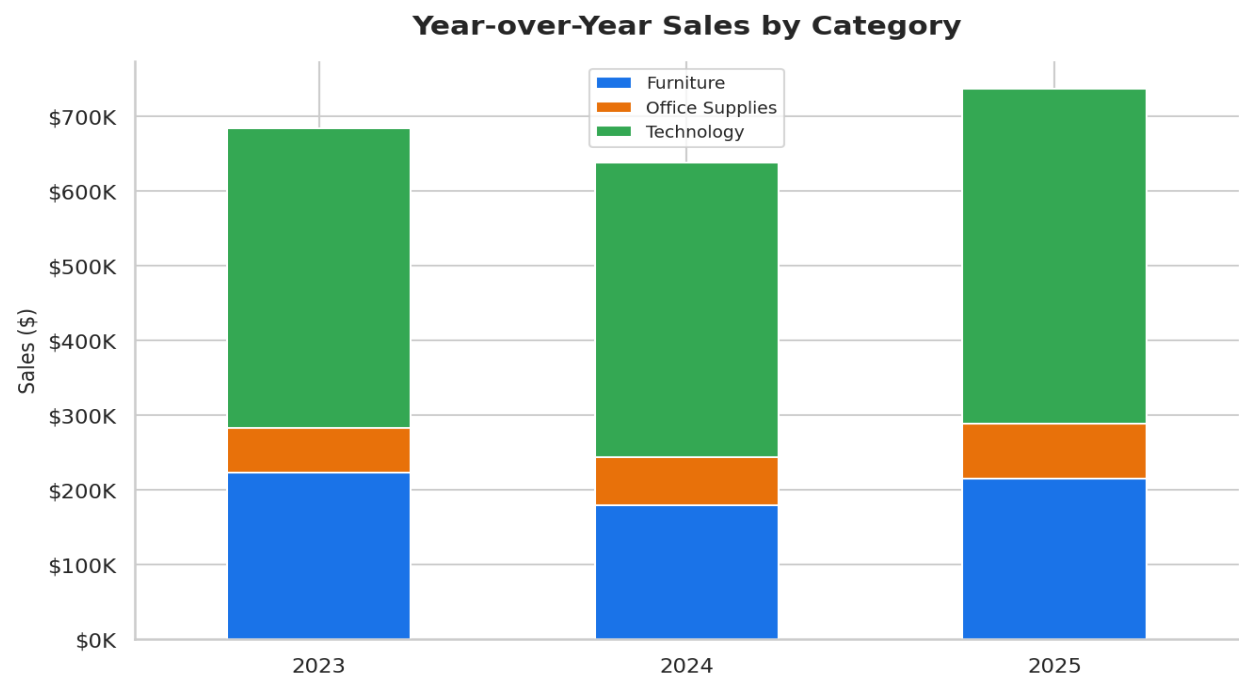
Sales Contribution by Customer Segment



■ **Key Takeaway — Consumer Segment is the Core Revenue Driver**

The Consumer segment contributes 52.6% of total sales, more than Corporate and Home Office combined. Retention and loyalty programs targeted at consumers will have the highest revenue impact.

6. Year-over-Year Sales by Category



■ **Key Takeaway — Consistent Category Mix Across Years**

Technology and Office Supplies have consistently outpaced Furniture every year. Furniture's slower contribution suggests it may need a pricing or marketing refresh to better compete for the unit volume increase.

Summary Storyboard — Business Narrative

Putting the six charts together tells a clear story: **the business is growing steadily, led by the West region and the Consumer segment, with Technology as the star category — but profitability is being quietly undermined by aggressive discounting.**

The Story in 4 Acts

Act	Insight	Business Implication
1. Growth	Sales trending upward with strong Q4 seasonality	Plan inventory & campaigns ahead of Q4
2. Strength	Technology sub-category & West region lead performance	Double down on what's working
3. Gap	South region and Furniture category underperform	Investigate root cause: pricing, demand, or reach
4. Risk	Discounts >30% turn profit negative	Cap discount policy to protect margin

Recommended Actions

- ✓ Cap discounts at 20-25% across all categories to safeguard profit margins.
- ✓ Increase marketing investment in the South region to close the performance gap.
- ✓ Maintain Technology category leadership through continued stock prioritization.
- ✓ Launch a Q3 pre-season campaign to extend the Q4 sales peak earlier in the year.
- ✓ Introduce a loyalty/rewards program targeting the high-value Consumer segment.